

Universidad Carlos III de Madrid

Aerospace Engineering

Computer Laboratory Report 2

Particle on Oscillating Loop

Mechanics Applied to Aerospace Engineering

Laboratory Group LA

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October 21, 2026

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1 Introduction

A small bead P of mass m slides without friction along a circular wire of centre C and radius R. The wire stands in the vertical plane Oxy, and the whole plane is turned back and forth about the vertical axis Oy with the angular velocity $\boldsymbol{\Omega}(t) = A\omega \sin(\omega t)\, \boldsymbol{j}$, where A and ω are known positive constants (Figure 1). The position of the bead on the wire is the angle φ between the axis Ox and the vector $\overrightarrow{CP}$ [1].

Gravity makes the bead swing around the wire like a pendulum, while the turning plane drags it sideways. The aim is to see how these two effects combine: which forces the wire has to make, why the mechanical energy changes, what the path looks like from the ground, when the bead goes over the top of a fixed loop, and how far a linearized equation can be trusted. Section 2 builds the model and the code, Section 3 explains each simulation next to its figure, Section 4 collects the points that concern the whole model and Section 5 sums up.

2 Methodology

2.1 Kinematics and degree of freedom

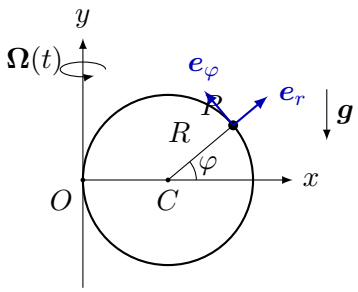


Figure 1: Geometry of the problem. The plane Oxy turns about Oy; $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ points out of the page. The polar basis $\{\boldsymbol{e}_r, \boldsymbol{e}_\varphi\}$ moves with the bead.

Two reference frames are used. $Oxyz$ turns with the plane of the wire: $\boldsymbol{i}$ is horizontal along the diameter through C, $\boldsymbol{j}$ points up along the axis and $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ is normal to the plane. $Ox_1y_1z_1$ is fixed to the ground and coincides with $Oxyz$ at $t = 0$. In the figure the axis Oy touches the wire at O, so $\overrightarrow{OC} = R\,\boldsymbol{i}$.

Degree of freedom. A free particle needs three coordinates. The wire takes two of them away: the bead must stay in the plane Oxy and at a distance R from C. One degree of freedom is left, and φ describes it. Since the plane turns with a given law, the position of the bead seen from the ground depends on φ and also directly on t.

How the plane turns. The angle θ turned by the plane grows at the rate $\dot\theta = \Omega$, starting from $\theta(0) = 0$. Integrating with the change $u = \omega\tau$, $\mathrm{d}u = \omega\,\mathrm{d}\tau$, and Barrow's rule,

$$\theta(t) = \int_0^t A\omega \sin(\omega\tau)\,\mathrm{d}\tau = A\int_0^{\omega t} \sin u\,\mathrm{d}u = A\big[-\cos u\big]_0^{\omega t} = A(1-\cos\omega t). \tag{1}$$

So the plane swings between $\theta = 0$ and $\theta = 2A$. With $A = \pi$ it makes one full turn by $t = T/2$ and then turns back, where $T = 2\pi/\omega$ is the period of the plane. The two bases are related by a rotation of angle θ about the vertical axis. The columns of

$$[{}_1R_0] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \tag{2}$$

are the components of $\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$ in $\{\boldsymbol{i}_1, \boldsymbol{j}_1, \boldsymbol{k}_1\}$. They are unit vectors perpendicular to each other, so $[{}_1R_0][{}_1R_0]^T = I$, and $\det[{}_1R_0] = \cos^2\theta + \sin^2\theta = 1$. As a check, differentiating $\boldsymbol{i} = \cos\theta\,\boldsymbol{i}_1 - \sin\theta\,\boldsymbol{k}_1$ gives $\mathrm{d}\boldsymbol{i}/\mathrm{d}t = -\dot\theta(\sin\theta\,\boldsymbol{i}_1 + \cos\theta\,\boldsymbol{k}_1) = -\Omega\,\boldsymbol{k}$, which is $\boldsymbol{\Omega} \times \boldsymbol{i}$, as it should be.

Motion along the wire. With the polar unit vectors $\boldsymbol{e}_r = \cos\varphi\,\boldsymbol{i} + \sin\varphi\,\boldsymbol{j}$ and $\boldsymbol{e}_\varphi = -\sin\varphi\,\boldsymbol{i} + \cos\varphi\,\boldsymbol{j}$, the position of the bead is

$$\boldsymbol{r} = \overrightarrow{OC} + \overrightarrow{CP} = R(1+\cos\varphi)\,\boldsymbol{i} + R\sin\varphi\,\boldsymbol{j}, \qquad x \equiv R(1+\cos\varphi), \tag{3}$$

where x is its distance to the axis of rotation. Seen from $Oxyz$ only φ changes. The chain rule gives $\frac{\mathrm{d}}{\mathrm{d}t}[R(1+\cos\varphi)] = -R\dot\varphi\sin\varphi$ and $\frac{\mathrm{d}}{\mathrm{d}t}[R\sin\varphi] = R\dot\varphi\cos\varphi$, so

$$\boldsymbol{v}_{\mathrm{rel}} = -R\dot\varphi\sin\varphi\,\boldsymbol{i} + R\dot\varphi\cos\varphi\,\boldsymbol{j} = R\dot\varphi\,\boldsymbol{e}_\varphi. \tag{4}$$

Differentiating again with the product rule, $\frac{\mathrm{d}}{\mathrm{d}t}(-R\dot\varphi\sin\varphi) = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)$ and $\frac{\mathrm{d}}{\mathrm{d}t}(R\dot\varphi\cos\varphi) = R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)$:

$$\boldsymbol{a}_{\mathrm{rel}} = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)\,\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} = R\ddot\varphi\,\boldsymbol{e}_\varphi - R\dot\varphi^2\,\boldsymbol{e}_r. \tag{5}$$

The first term speeds the bead up along the wire, and the second one is the centripetal acceleration towards C.

Motion seen from the ground. O lies on the axis and never moves, so the velocity and acceleration in the fixed frame are $\boldsymbol{v} = \boldsymbol{v}_{\mathrm{rel}} + \boldsymbol{\Omega}\times\boldsymbol{r}$ and $\boldsymbol{a} = \boldsymbol{a}_{\mathrm{rel}} + \dot{\boldsymbol{\Omega}}\times\boldsymbol{r} + \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) + 2\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}}$, with $\Omega = A\omega\sin\omega t$ and $\dot\Omega = A\omega^2\cos\omega t$. With $\boldsymbol{j}\times\boldsymbol{i} = -\boldsymbol{k}$, $\boldsymbol{j}\times\boldsymbol{j} = \boldsymbol{0}$ and $\boldsymbol{j}\times\boldsymbol{k} = \boldsymbol{i}$, the four extra terms are

$$\boldsymbol{\Omega}\times\boldsymbol{r} = \Omega\,x\,(\boldsymbol{j}\times\boldsymbol{i}) + \Omega R\sin\varphi\,(\boldsymbol{j}\times\boldsymbol{j}) = -\Omega x\,\boldsymbol{k}, \qquad \dot{\boldsymbol{\Omega}}\times\boldsymbol{r} = -\dot\Omega x\,\boldsymbol{k}, \tag{6}$$

$$\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = \Omega\,\boldsymbol{j}\times(-\Omega x\,\boldsymbol{k}) = -\Omega^2 x\,(\boldsymbol{j}\times\boldsymbol{k}) = -\Omega^2 x\,\boldsymbol{i}, \tag{7}$$

$$2\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}} = 2\Omega R\dot\varphi\big[-\sin\varphi\,(\boldsymbol{j}\times\boldsymbol{i}) + \cos\varphi\,(\boldsymbol{j}\times\boldsymbol{j})\big] = 2\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k}. \tag{8}$$

Adding them,

$$\boldsymbol{v} = R\dot\varphi\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}, \tag{9}$$

$$\boldsymbol{a} = \big[-R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) - \Omega^2 x\big]\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} + \big[-\dot\Omega x + 2\Omega R\dot\varphi\sin\varphi\big]\boldsymbol{k}. \tag{10}$$

Since $\boldsymbol{e}_\varphi \perp \boldsymbol{k}$, the speed is $|\boldsymbol{v}|^2 = R^2\dot\varphi^2 + \Omega^2 x^2$: the speed along the wire plus the speed at which the plane carries the bead around the axis.

2.2 Forces and equation of motion

Only two real forces act on the bead: its weight and the push of the wire. The wire is smooth, so it cannot push along its own tangent $\boldsymbol{e}_\varphi$:

$$\boldsymbol{W} = -mg\,\boldsymbol{j} = -mg\sin\varphi\,\boldsymbol{e}_r - mg\cos\varphi\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{N} = N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k}, \tag{11}$$

where $\boldsymbol{j} = \sin\varphi\,\boldsymbol{e}_r + \cos\varphi\,\boldsymbol{e}_\varphi$ was used. To write Newton's law in the turning frame the inertia forces must be added. From Equations (6) to (8),

$$\boldsymbol{F}_{\mathrm{cf}} = -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = m\Omega^2 x\,\boldsymbol{i} = mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\big[\cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi\big], \tag{12}$$

$$\boldsymbol{F}_{\mathrm{Eul}} = -m\dot{\boldsymbol{\Omega}}\times\boldsymbol{r} = m\dot\Omega x\,\boldsymbol{k} = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\boldsymbol{k}, \tag{13}$$

$$\boldsymbol{F}_{\mathrm{Cor}} = -2m\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}} = -2m\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k} = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\boldsymbol{k}, \tag{14}$$

with $\boldsymbol{i} = \cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi$. In $Oxyz$ Newton's second law reads $m\boldsymbol{a}_{\mathrm{rel}} = \boldsymbol{W} + \boldsymbol{N} + \boldsymbol{F}_{\mathrm{cf}} + \boldsymbol{F}_{\mathrm{Eul}} + \boldsymbol{F}_{\mathrm{Cor}}$. The reaction has no component along $\boldsymbol{e}_\varphi$, and the Euler and Coriolis forces point along $\boldsymbol{k}$, so the component along $\boldsymbol{e}_\varphi$ contains no unknown force: $mR\ddot\varphi = -mg\cos\varphi - m\Omega^2 x\sin\varphi$, that is,

$$\ddot\varphi = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi. \tag{15}$$

The other two components give the reaction. Along $\boldsymbol{e}_r$, $-mR\dot\varphi^2 = N_r - mg\sin\varphi + m\Omega^2 x\cos\varphi$, and along $\boldsymbol{k}$, $0 = N_z + m\dot\Omega x - 2m\Omega R\dot\varphi\sin\varphi$. Solving for the reaction,

$$N_r = -mR\dot\varphi^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{16}$$

$$N_z = mA\omega R\big[2\dot\varphi\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{17}$$

A few things can be read directly from Equation (15):

- With the plane at rest ($A = 0$) it is a pendulum, $\ddot{\varphi} = -(g/R)\cos\varphi$ (a cosine because φ is measured from the horizontal), with a stable equilibrium at the bottom, $\varphi = -\pi/2$, and an unstable one at the top, $\varphi = \pi/2$.
- Only the centrifugal force changes the motion along the wire; the Euler and Coriolis forces are normal to the plane and only load the wire through N_z. For item 5 the centrifugal term is at most $(A\omega)^2/(g/R) \approx 1\%$ of gravity, so the bead should move almost like a pendulum on a fixed loop.
- The bead cannot leave the wire, so $\boldsymbol{N}$ can have any sign: unlike the string of Lab 1, nothing can stop the model.

2.3 Mechanical energy

The kinetic energy uses the velocity seen from the ground, Equation (9):

$$E_k = \tfrac{1}{2}m(R^2\dot{\varphi}^2 + \Omega^2 x^2), \qquad E_p = mgR\sin\varphi, \qquad E = E_k + E_p. \tag{18}$$

The weight is already inside E, so E only changes through the work of the reaction. Using Equations (9) and (11) and $\boldsymbol{e}_r \cdot \boldsymbol{e}_\varphi = 0$,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \boldsymbol{N}\cdot\boldsymbol{v} = (N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k})\cdot(R\dot{\varphi}\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}) = -\Omega\,x\,N_z. \tag{19}$$

On a fixed wire the reaction is always perpendicular to the velocity and does no work. Here the wire moves with the plane and pushes the bead along $\boldsymbol{k}$, so it does work whenever $\Omega \neq 0$. E is conserved only for $A = 0$.

2.4 Fixed loop: when does the bead go over the top?

For $A = 0$ the energy is conserved, and that is enough to answer the question. Multiplying Equation (15) by $\dot{\varphi}$ gives $\dot{\varphi}\ddot{\varphi} + (g/R)\cos\varphi\,\dot{\varphi} = 0$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}\dot{\varphi}^2) = \dot{\varphi}\ddot{\varphi}$ and $\frac{\mathrm{d}}{\mathrm{d}t}\sin\varphi = \cos\varphi\,\dot{\varphi}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\tfrac{1}{2}\dot{\varphi}^2 + \frac{g}{R}\sin\varphi\Big] = 0 \quad\Rightarrow\quad \tfrac{1}{2}\dot{\varphi}^2 + \frac{g}{R}\sin\varphi = \tfrac{1}{2}\dot{\varphi}_0^2 + \frac{g}{R}\sin\varphi_0, \tag{20}$$

which is $E/(mR^2) =$ constant. The potential energy is highest at the top, $\varphi = \pi/2$, where $E_p = mgR$. If the bead still has some speed when it gets there, it goes over and φ keeps growing; if not, it stops before the top and comes back. Asking for $\dot{\varphi}^2 > 0$ at $\varphi = \pi/2$ in Equation (20) gives the criterion

$$|\dot{\varphi}_0| > \dot{\varphi}_c = \sqrt{\frac{2g}{R}(1 - \sin\varphi_0)}. \tag{21}$$

When both sides are equal the bead is on the separatrix $\dot{\varphi} = \pm\sqrt{(2g/R)(1-\sin\varphi)}$: it creeps up to the top but needs an infinite time to get there. Below the separatrix φ oscillates between the two angles where $\dot{\varphi} = 0$, given by $\sin\varphi = \sin\varphi_0 + R\dot{\varphi}_0^2/(2g)$. Table 1 applies the criterion to the nine initial conditions of item 7.

Table 1: Critical rate of Equation (21) ($g/R = 9.81\,\mathrm{s}^{-2}$) and predicted motion for the nine initial conditions.

φ_0 (rad)	$\dot{\varphi}_c$ (rad/s)	$\dot{\varphi}_0 = 0$	$\dot{\varphi}_0 = 1.5$ rad/s	$\dot{\varphi}_0 = 3$ rad/s
0	4.429	oscillates	oscillates	oscillates
$\pi/4$	2.397	oscillates	oscillates	goes over the top
$\pi/2$	0	rest at the top (unstable)	goes over the top	goes over the top

2.5 Linearized equation

Now the bead stays close to the bottom, $\varphi = \delta - \pi/2$ with $|\delta| \ll 1$, and the rotation is weak, $A^2 \ll g/(R\omega^2)$. Then $\cos\varphi = \cos(\delta - \pi/2) = \sin\delta \approx \delta$, $\sin\varphi = -\cos\delta \approx -1$, $1 + \cos\varphi \approx 1 + \delta$ and $\ddot{\varphi} = \ddot{\delta}$. Substituting in Equation (15),

$$\ddot{\delta} = -\frac{g}{R}\,\delta + \Omega^2(1+\delta) = -\frac{g}{R}\,\delta + \Omega^2 + \Omega^2\delta. \tag{22}$$

The hypothesis means $\Omega^2 \le A^2\omega^2 \ll g/R$, so $\Omega^2\delta$ is small next to $(g/R)\delta$ and is dropped. With $\sin^2(\omega t) = \frac{1}{2}[1-\cos(2\omega t)]$,

$$\ddot{\delta} + \omega_n^2\,\delta = \Omega^2 = F_0[1-\cos(2\omega t)], \qquad \omega_n = \sqrt{g/R}, \qquad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{23}$$

This is an undamped oscillator of natural frequency ω_n, pushed at the frequency 2ω [2]. The push is the centrifugal force: it is zero when the plane stops and largest when it turns fastest, which happens twice per period of the plane. The statement gives no initial conditions for item 8, and neither does the problem sheet [3]. We take the bead at rest at the bottom, $\delta(0) = 0$ and $\dot{\delta}(0) = 0$, so that all its motion comes from the loop. The particular solution is a constant F_0/ω_n^2 plus $C\cos(2\omega t)$; substituting the second one, $(\omega_n^2 - 4\omega^2)C = -F_0$. Adding the free solution $a\cos\omega_n t + b\sin\omega_n t$ and imposing the initial conditions ($\dot{\delta}(0) = b\,\omega_n = 0$),

$$\delta(t) = \frac{F_0}{\omega_n^2} - \frac{F_0\cos(2\omega t)}{\omega_n^2 - 4\omega^2} + \Big(\frac{F_0}{\omega_n^2 - 4\omega^2} - \frac{F_0}{\omega_n^2}\Big)\cos(\omega_n t), \qquad 2\omega \neq \omega_n. \tag{24}$$

So the bead oscillates about $\delta_s = F_0/\omega_n^2 = A^2R\omega^2/(2g)$, a little outwards from the bottom, because the centrifugal force pushes it away from the axis. If $2\omega = \omega_n$ the denominator of Equation (24) is zero. This is resonance: each push arrives in step with the natural swing. The particular solution becomes $-\frac{F_0}{2\omega_n}t\sin(\omega_n t)$, which satisfies $\ddot{\delta} + \omega_n^2\delta = -F_0\cos(\omega_n t)$, and with the same initial conditions

$$\delta(t) = \frac{F_0}{\omega_n^2}(1-\cos\omega_n t) - \frac{F_0}{2\omega_n}\,t\sin(\omega_n t), \qquad 2\omega = \omega_n. \tag{25}$$

The amplitude now grows linearly in time, so sooner or later $\delta \ll 1$ stops being true, and the linearization with it.

2.6 Numerical method

`ode45` integrates first order systems [4], so Equation (15) is written for the state vector $\boldsymbol{X} = [\varphi, \dot{\varphi}]^T$:

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - \Omega(t)^2\,(1+\cos X_1)\sin X_1 \end{pmatrix}, \qquad \boldsymbol{X}(0) = \begin{pmatrix}\varphi_0 \\ \dot{\varphi}_0\end{pmatrix}, \qquad \Omega(t) = A\omega\sin(\omega t). \tag{26}$$

This system is the MATLAB function `diffeq(t,X,g,w,R,A)`, and Equation (23) is `diffeqlin`, with the state $[\delta, \dot{\delta}]^T$. The tolerances are 10^{-8} (relative) and 10^{-10} (absolute), and in item 5 the solution is stored at 10 000 instants of $[0, 3T]$, about 125 per swing [1]. Forces and energies are computed from the stored φ and $\dot{\varphi}$ with exactly Equations (12) to (14) and (16) to (18). To check Equation (19) without adding a state, the work of the wire is computed from the power $P_i = -\Omega_i x_i N_{z,i}$ with the trapezoidal rule,

$$W_n = \sum_{i=1}^{n-1} \tfrac{1}{2}\,(P_i + P_{i+1})(t_{i+1} - t_i), \tag{27}$$

and compared with $E - E_0$. The position in the fixed frame comes from Equations (1) and (2): $x_1 = x\cos\theta$, $y_1 = R\sin\varphi$, $z_1 = -x\sin\theta$. In all the cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$. Items 5 and 6 use $A = \pi$, $R = 1\,\mathrm{m}$, $\omega = 0.1\,\mathrm{rad/s}$ and $\varphi_0 = \dot{\varphi}_0 = 0$ over $[0, 3T]$, with $T = 62.83\,\mathrm{s}$. Item 7 uses $A = 0$, $R = 1\,\mathrm{m}$, the conditions of Table 1 and $\varphi_0 = \pi/4$ with $\dot{\varphi}_0 = (1 \pm 0.01)\,\dot{\varphi}_c$, over 4 s. Item 8 uses $A = 0.1$, $R = 0.01\,\mathrm{m}$, $\omega = \omega_n,\ \omega_n/2,\ \omega_n/4$ with $\omega_n = 31.32\,\mathrm{rad/s}$, and $\delta_0 = \dot{\delta}_0 = 0$, over 10 s.

3 Results

3.1 Oscillating loop: angle, reaction and inertia forces

At $t = 0$ the plane is at rest, and Equation (15) gives $\ddot{\varphi}(0) = -g/R = -9.81\,\mathrm{rad/s^2}$, as the code prints. The bead falls from the end of the horizontal diameter and swings between $\varphi = 0$ and $\varphi = -\pi$ (Figure 2a,b); at $\varphi = -\pi$ it passes through O, on the axis. One swing takes 2.37 s, longer than the small-oscillation period $2\pi\sqrt{R/g} = 2.006\,\mathrm{s}$, because an amplitude of 90° around the bottom is far from small. Over the three periods of the plane the amplitude does not change, as expected from a

centrifugal term of only 1% of gravity. The highest angle reached is $\varphi = 0.0118\,\text{rad}$: the energy given by the wire lifts the bead about 1 cm above its starting point.

The radial reaction (Figure 2c) is zero at the turning points and reaches $-29.43\,\text{N}$, that is $-3mg$, at the bottom. There Equation (20) gives $\dot{\varphi}^2 = 2g/R$, and Equation (16) gives $N_r = -2mg - mg$: the wire holds the weight and also pulls the bead towards C to make it turn. The normal reaction (Figure 2d) reaches $\pm 2.78\,\text{N}$, and almost all of it balances the Coriolis force (Figure 2f), whose largest value is $2mA\omega R\dot{\varphi}_{\max} = 2\pi(0.1)(4.43) = 2.78\,\text{N}$. It is large because the bead runs fast along the wire ($\dot{\varphi}$ up to $4.4\,\text{rad/s}$) while the plane turns, and it almost disappears at $t = 0, T/2, T, \ldots$, when the plane stops to turn back. The centrifugal force (Figure 2e) always points away from the axis and peaks at $m(A\omega)^2(2R) = 0.197\,\text{N}$ when the plane turns fastest, at $t = T/4, 3T/4, \ldots$ The Euler force stays below $0.063\,\text{N}$, because the plane changes its rate slowly ($|\dot{\Omega}| \le A\omega^2 = 0.031\,\text{rad/s}^2$).

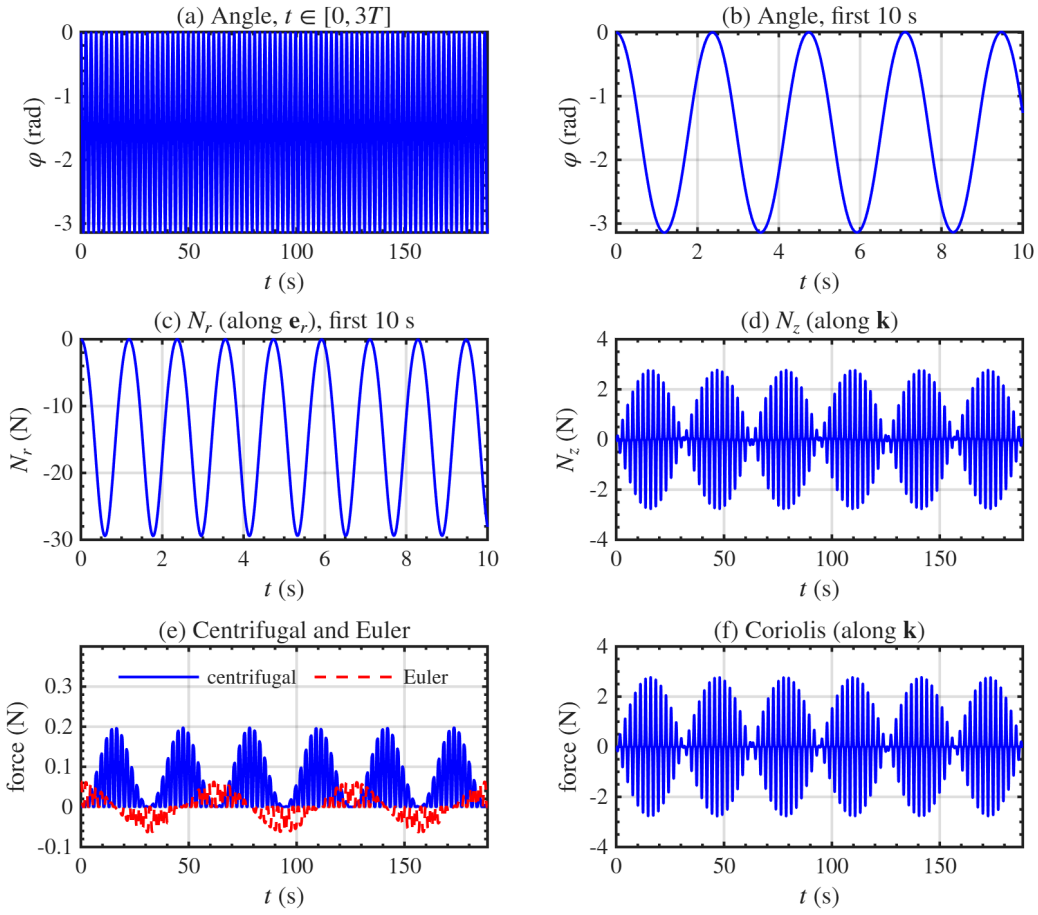


Figure 2: Oscillating loop, item 5. (a) Angle in $[0, 3T]$ and (b) in the first 10 s. (c) Reaction along $\boldsymbol{e}_r$, first 10 s. (d) Reaction along $\boldsymbol{k}$. (e) Centrifugal ($\boldsymbol{i}$) and Euler ($\boldsymbol{k}$) forces. (f) Coriolis force ($\boldsymbol{k}$).

3.2 Energy

In every swing E_k and E_p exchange about 9.8 J (Figure 3a), as in any pendulum. Their sum is not constant (Figure 3b): $E - E_0$ moves between $-0.082\,\text{J}$ and $0.313\,\text{J}$, oscillating with each swing inside an envelope that follows Ω^2. While the plane speeds up, the wire pushes the bead along with it and gives it energy; while the plane slows down, the wire takes that energy back. The plane changes its speed slowly compared with a swing ($T = 62.8\,\text{s}$ against $2.37\,\text{s}$), so the energy comes back to E_0 each time the plane stops, and after $3T$ the net gain is only $0.0006\,\text{J}$. This is why the amplitude in Figure 2a does not drift.

The work of the wire from Equation (27) falls on top of $E - E_0$; the largest difference, $1.5 \times 10^{-4}\,\text{J}$ (0.05% of the range of E), comes from the trapezoidal rule. This confirms Equation (19) and the expressions of N_z and E in the code. Tighter tolerances (10^{-11}, 10^{-12}) change φ by at most $1.4 \times 10^{-4}\,\text{rad}$.

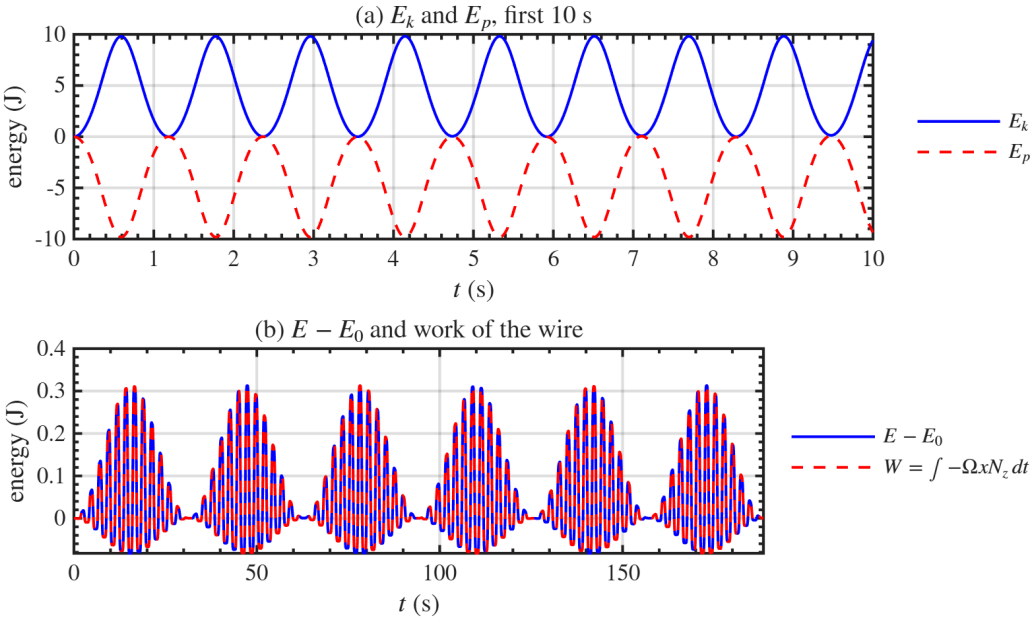


Figure 3: Energy, item 5. (a) E_k and E_p in the first 10 s (the pattern repeats over $[0, 3T]$). (b) $E - E_0$ and work of the wire, Equation (27), in $[0, 3T]$.

3.3 Trajectory in the fixed frame

Seen from the ground, the bead always stays on the surface swept by the wire as it turns about Oy: a torus without a hole, because the circle touches the axis at O (Figure 4a). From above (Figure 4b) each swing draws one petal: the bead goes out to $x = 2R$ when $\varphi = 0$ and back to the axis when $\varphi = -\pi$. The petals spread around the axis because the plane turns to $\theta = 2\pi$ at $t = T/2$ and back to 0 at $t = T$ (Equation (1)). The three periods draw almost the same flower; the curves do not match exactly because 2.37 s does not divide T. As a check, $\sqrt{x_1^2 + z_1^2}$ equals x to 2×10^{-16} m, since a rotation about the axis cannot change the distance to it.

The script also writes an animation of the first period, `trajectory_P.avi`, with the wire turned by $\theta(t)$ and about 20 frames per swing. Each frame is printed to a fixed size with `print(gcf,'-RGBImage')`, because `getframe` [1] sometimes returned frames of different sizes, which `VideoWriter` rejects.

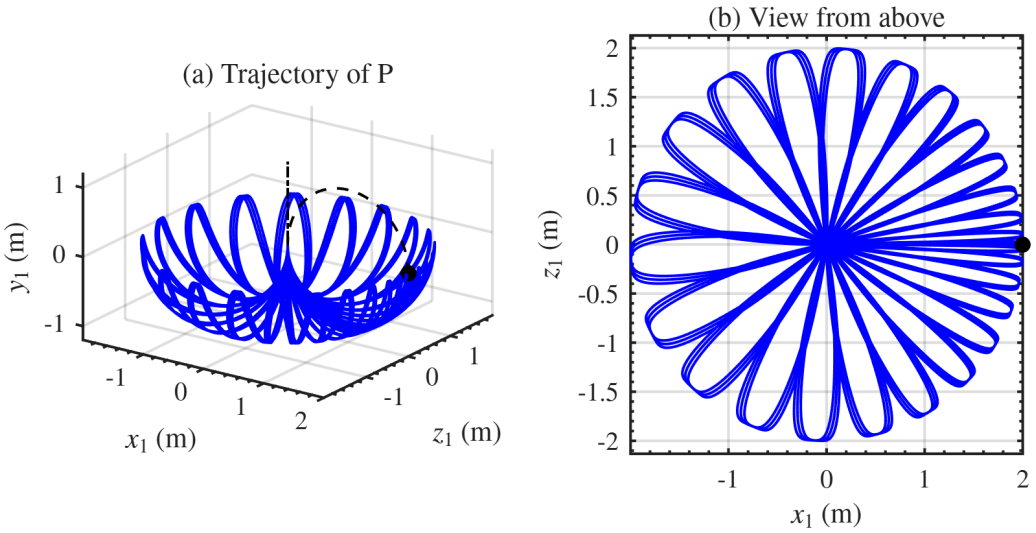


Figure 4: Path of P in the fixed frame for $t \in [0, 3T]$. (a) 3D view with the wire at $t = 0$ (dashed), the axis (dash-dot) and the start (dot). (b) View from above.

3.4 Fixed loop: phase portrait

Figure 5a shows the nine cases of Table 1 with the separatrix of Equation (21). Five of them, $(0, 0)$, $(0, 1.5)$, $(0, 3)$, $(\pi/4, 0)$ and $(\pi/4, 1.5)$, are closed curves around the stable equilibrium $(-\pi/2, 0)$, so the bead oscillates. The period grows from 2.37 s to 3.36 s as the curve gets closer to the separatrix, because the bead slows down near the top. The cases $(\pi/4, 3)$, $(\pi/2, 1.5)$ and $(\pi/2, 3)$ start above the separatrix and give open curves: the bead goes round and round, slowest at the top and fastest at the bottom. The case $(\pi/2, 0)$ sits on the unstable equilibrium and stays there as a single point; numerically $\cos(\pi/2) \approx 6 \times 10^{-17}$, far too small to move it in 4 s. All nine cases behave as Table 1 predicts.

To test the threshold itself, $\varphi_0 = \pi/4$ was also run with $\dot\varphi_0$ 1% below and 1% above $\dot\varphi_c$ (Figure 5b). Both runs climb towards the top and hang there for almost a second, which shows how slowly the bead moves near the separatrix. The first one reaches $\varphi = 1.463$ rad, just short of $\pi/2$, and falls back; the second one goes over. In all the runs Equation (20) holds to better than $2 \times 10^{-7}\, mgR$.

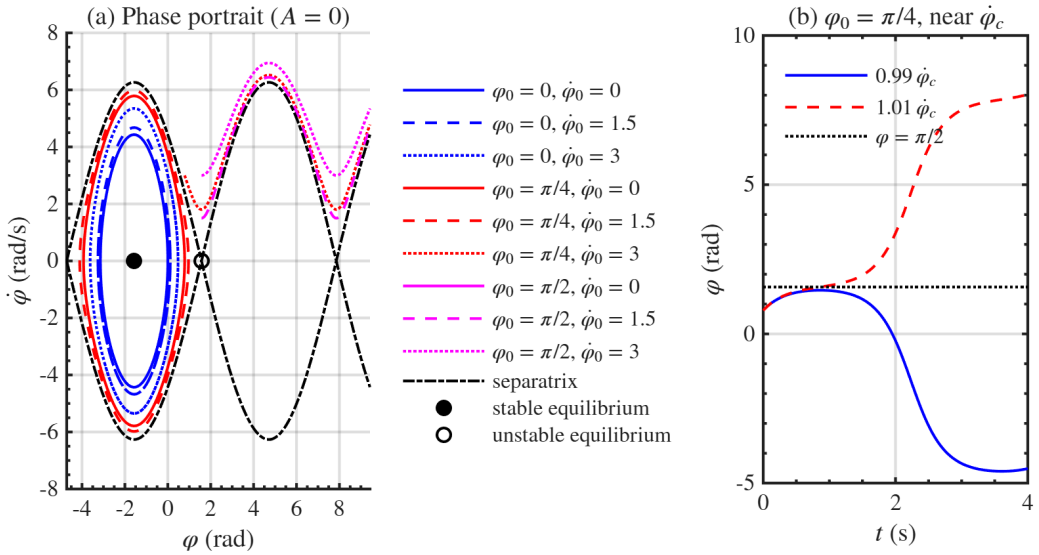


Figure 5: Fixed loop ($A = 0$), item 7. (a) Phase portrait for the nine initial conditions, with the separatrix and the two equilibria. (b) $\varphi_0 = \pi/4$ with $\dot\varphi_0$ 1% below and above the critical rate of Equation (21).

3.5 Linearized and full equation

Figure 6a shows the percentage error $100\,|\delta_{\text{lin}} - \delta_{\text{full}}| / \max|\delta_{\text{full}}|$. It is divided by the amplitude of δ and not by δ itself, because δ crosses zero and the error would blow up there. The `ode45` solution of Equation (23) matches the closed forms Equations (24) and (25) to 2×10^{-8} rad, so every difference in the figure comes from the model and not from the integration. Table 2 gives the numbers, and each frequency shows a different way in which the linearization can fail.

Table 2: Linearized against full equation, $t \in [0, 10]$ s, with $\omega_n = \sqrt{g/R}$.

ω	$\max\lvert\delta_{\text{full}}\rvert$ (rad)	error at $t = 5$ s	largest error
ω_n	0.0134	6.68%	21.4%
$\omega_n/2$	0.196	3.30%	27.8%
$\omega_n/4$	0.0008	0.32%	1.86%

For $\omega = \omega_n$, δ stays small but the error grows steadily up to 21%. The cause is the term $\Omega^2\delta$ dropped in Section 2.5. It lowers the stiffness of the full equation from $g/R = 981\,\text{s}^{-2}$ by about $A^2\omega^2/2 = 4.9\,\text{s}^{-2}$ on average, so the full solution oscillates 0.25% slower and falls behind the linear one, by about 0.8 rad of phase after 10 s.

For $\omega = \omega_n/2$ the forcing frequency 2ω equals ω_n: this is the resonance of Equation (25) [2]. Both solutions grow along the envelope $F_0 t/(2\omega_n)$ (Figure 6b) and reach 0.196 rad (11°) after 10 s. At first

they agree, but as δ grows, $\sin\delta \approx \delta$ gets worse, the full pendulum slows down with the amplitude and drifts out of step with the forcing, and the error reaches 27.8%.

For $\omega = \omega_n/4$ the forcing is far from resonance and 16 times weaker than in the first case. δ stays below 8×10^{-4} rad, the dropped term is negligible and the error stays below 2%. This is the only case of the three where the linearization really works.

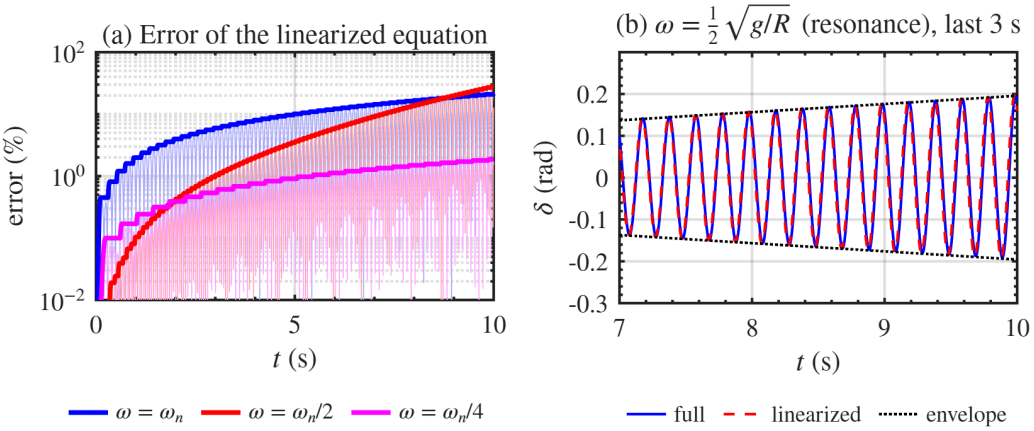


Figure 6: Item 8, $R = 0.01$ m, $A = 0.1$, $\omega_n = \sqrt{g/R}$. (a) Percentage error of the linearized equation; light lines: error, thick lines: largest error up to each time. (b) Full and linearized solutions at resonance in the last 3 s, with the envelope $\pm F_0 t/(2\omega_n)$ of Equation (25).

4 Discussion

Two points concern the whole model rather than one figure. First, energy. In the fixed frame it is not conserved because the wire moves (Equation (19)). In the turning frame the reaction, the Euler force and the Coriolis force do no work, since they are perpendicular to $\boldsymbol{v}_{\text{rel}}$, but the centrifugal potential $-\frac{1}{2}m\Omega^2x^2$ changes with time, so $\frac{1}{2}mR^2\dot{\varphi}^2 + E_p - \frac{1}{2}m\Omega^2x^2$ is not conserved either. Only a constant Ω (in particular $A = 0$) gives a conserved energy, which item 7 uses.

Second, the initial conditions of item 8, which we had to choose. Starting with $\delta(0) \neq 0$ would add a free oscillation and change the size of the errors, but not the two mechanisms behind them, the phase lag and the resonance.

5 Conclusions

The bead on an oscillating loop has one degree of freedom, φ, and in the turning frame its equation of motion contains only gravity and the centrifugal force; the wire balances the Euler and Coriolis forces. For the data of item 5 the plane turns slowly and gently, so the bead swings like a pendulum of 90° amplitude with a period of 2.37 s. The wire supplies up to $3mg$ in its plane and up to 2.78 N across it, almost all against the Coriolis force. That push does work, so the mechanical energy is not conserved, but the work cancels out every time the plane stops.

With the loop fixed, energy conservation gives the critical rate $\dot{\varphi}_c = \sqrt{(2g/R)(1 - \sin\varphi_0)}$, and all the simulated cases follow it. The linearized equation is a forced oscillator: it works when the forcing is weak and far from resonance, drifts in phase when the dropped term $\Omega^2\delta$ is not small enough, and fails at resonance, $\omega = \omega_n/2$, where the amplitude grows until $\delta \ll 1$ is no longer true.

References

[1] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: Laboratory Guidelines*, 2026.

[2] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: course notes*, Sec. 5.4.

[3] Aerospace Engineering Department (UC3M), *Particle exercises and problems*, Problem 34.

[4] The MathWorks, Inc., *MATLAB documentation: ode45, odeset, VideoWriter, print.*